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Multi-Mode Rolling-Encoded Ultrasound

Abstract

Systems and methods for a multi-mode rolling-encoded ultrasound are described. These systems and methods include an ultrasound device that provides an encoded ultrasound signal, which includes one or more multi-mode waveforms that contain multiple mode types, multiple variants from a single mode type, or a mixture of such. The encoded ultrasound signals are used to image or otherwise gather information from a target during an ultrasound scan, such as a portion of a patient anatomy. The reflected encoded signals are received and decoded, resulting in an increased resolution and frame rate. This allows for more-efficient operation and resource utilization.

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Background/Summary

BACKGROUND

[0001] Ultrasound systems can generate ultrasound images by transmitting sound waves at frequencies above the audible spectrum into a body, receiving echo signals caused by the sound waves reflecting from internal body parts, and converting the echo signals into electrical signals for image generation. Because they are non-invasive and can provide immediate imaging results, ultrasound systems are used ubiquitously. In ultrasound imaging, less artifacts, better resolution and penetration, and faster frame rate are always desired. However, in real applications, because of the limitations due to physics, safety, etc., often there is a compromise scenario involving unwanted artifacts, unideal resolution, inadequate penetration, or a low frame rate, to name a few.

[0002] In certain ultrasound modes (e.g., triplex), there are obvious band artifacts in the pulsed-wave (PW) Doppler spectrum due to data discontinuity. In some applications, the resolution, penetration, frame rate, or any number of other factors may not be sufficient due to the patient type. Even with parameters at their current technological limit, it is always preferable to have these properties (e.g., frame rate, resolution, etc.) enhanced for more accurate information and better diagnosis results. Current methods are aimed at solving the artifact issue and/or improving the image properties (e.g., resolution, penetration, frame rate, etc.), but they all have drawbacks and trade-offs. For example, to address the artifact issue in PW spectrum, extrapolation is used to estimate the missing data. However, it is still an approximation, and cannot remove the artifacts completely. In another example, signal frequency can be lowered to provide better penetration, but this results in a loss of resolution.

[0003] A robust solution is needed to realize higher resolution and frame rate without the sacrifices and trade-offs endemic to current methodologies.

SUMMARY

[0004] Devices and methods for a multi-mode rolling-encoded ultrasound are described. These systems and methods include an ultrasound device that provides an encoded ultrasound signal, which includes one or more multi-mode waveforms that contain multiple mode types, multiple variants from a single mode type, or a mixture of such. The encoded ultrasound signals are used to image or otherwise gather information from a target during an ultrasound scan, such as a portion of a patient anatomy. The reflected encoded signals are received and decoded, resulting in an increased resolution and frame rate. This allows for more-efficient operation and resource utilization compared to conventional ultrasound systems.

[0005] In some implementations, an ultrasound device is disclosed. The ultrasound device includes an ultrasound scanner configured to generate an ultrasound signal, where the ultrasound signal includes a plurality of ultrasound signal modes and is configured for transmission by the ultrasound scanner at a subject. The ultrasound device further includes, in aspects, one or more processors and a memory. The memory stores instructions that, when executed by the one or more processors, cause the one or more processors to encode the ultrasound signal as a mixed waveform. The encoding is based on the plurality of ultrasound signal modes. According to some examples, the plurality of ultrasound signal modes includes two or more of an amplitude mode (“A-mode”), a brightness mode (“B-mode”), a motion mode (“M-mode”), a Doppler-based mode, etc., or different variants of these modes. In aspects, the instructions further cause the one or more processors to transmit the encoded ultrasound signal at the subject. The transmission, in aspects, is performed by the ultrasound scanner. In some examples, encoding the ultrasound signal includes combining the plurality of ultrasound signal modes based on one or more of a polarity, a time of transmission, or a location of transmission. In aspects, the plurality of ultrasound signal modes is combined using an orthogonal matrix operator. In some examples, at least one of the plurality of ultrasound signal modes is a non-image-based signal mode.

[0006] In some implementations, an ultrasound device is disclosed. The ultrasound device includes an ultrasound scanner, and the ultrasound scanner is configured to receive an encoded ultrasound

signal. The encoded ultrasound signal includes reflections of a plurality of ultrasound signal modes from a subject. The ultrasound device also includes one or more processors and a memory, where the memory stores instructions that, when executed by the one or more processors, cause the one or more processors to decode the received encoded ultrasound signal into ultrasound data. The ultrasound data includes the plurality of ultrasound signal modes. According to some examples, the plurality of ultrasound signal modes includes two or more of an A-mode, a B-mode, an M-mode, a Doppler-based mode, etc., or different variants of these modes. In aspects, the decoding of the received encoded ultrasound signal into the ultrasound data is based on one or more of a polarity, a time of transmission, or a location of transmission for the encoded ultrasound signal. According to some examples, the instructions further cause the one or more processors to produce an output based on the ultrasound data. The output can be an image of an anatomy of the subject, a determination of a speed of sound, a determination of a background noise, or a detection of one or more artifacts. According to some examples, the instructions further cause the one or more processors to generate a filtered ultrasound data based on the ultrasound data. In aspects, the decoding of the received encoded ultrasound signal into the ultrasound data includes processing the received encoded ultrasound signal using an orthogonal matrix operator, and the orthogonal matrix operator can be based on a Fourier transformation.

[0007] In some implementations, an ultrasound device is disclosed. The ultrasound device includes an ultrasound scanner, and the ultrasound scanner is configured to generate a first ultrasound signal. The first ultrasound signal includes a plurality of ultrasound signal modes. The first ultrasound signal is configured for transmission by the ultrasound scanner at a subject. The ultrasound scanner is further configured to receive an encoded ultrasound signal, and the encoded ultrasound signal includes reflections of the plurality of ultrasound signal modes from the subject. The ultrasound device further includes one or more processors and a memory. The memory stores instructions that, when executed by the one or more processors, cause the one or more processors to encode the first ultrasound signal as a mixed waveform. The encoding of the first ultrasound signal is based on the plurality of ultrasound signal modes. The one or more processors are further caused to, based on the received encoded ultrasound signal, decode the received encoded ultrasound signal into second ultrasound data. The second ultrasound data includes the plurality of ultrasound signal modes. In aspects, the plurality of ultrasound signal modes are two or more of an A-mode, a B-mode, an M-mode, a Doppler-based mode, or different variants of these modes. According to some examples, the encoding of the first ultrasound signal and the decoding of the received encoded ultrasound signal includes processing the plurality of ultrasound signal modes and the received encoded ultrasound signal, respectively, using an orthogonal matrix operator. In aspects, the instructions further cause the one or more processors to produce a filtered ultrasound data. The filtered ultrasound data is based on the second ultrasound signal, and the one or more processors are further caused to produce, based on the filtered ultrasound data, an output. In some examples, the output is an image of an anatomy of the subject, a determination of a speed of sound, a determination of a background noise, or a detection of one or more artifacts. According to some examples, the instructions further cause the one or more processors to transmit, by the ultrasound scanner, the first encoded ultrasound signal at the subject.

[0008] In some implementations, a method for encoding and decoding a multi-mode rolling-encoded ultrasound signals is disclosed. The method includes encoding, by one or more processors, a first ultrasound signal as a mixed waveform, the first ultrasound signal. The encoding is based on a plurality of ultrasound signal modes. The encoded first ultrasound signal includes the plurality of ultrasound signal modes and is configured for transmission by the ultrasound scanner at a subject. The method, in aspects, further includes generating, by an ultrasound scanner, the first ultrasound signal. In some examples, the method further includes receiving, by the ultrasound scanner and from the subject, an encoded second ultrasound signal. The encoded second ultrasound signal includes reflections of the plurality of ultrasound signal modes. The method, in aspects, further

includes decoding the received encoded second ultrasound signal into ultrasound data. The decoding is based on the received encoded second ultrasound signal, and the ultrasound data includes the plurality of ultrasound signal modes.

[0009] Other devices and methods to provide a multi-mode rolling-encoded ultrasound are also described. These other devices and methods, in addition to those already disclosed, can be combined to generate additional devices and methods, which, though not explicitly disclosed herein, still are the same in concept as the devices and methods described explicitly herein. The devices and methods explicitly outlined herein are meant to be illustrative and not limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The appended drawings illustrate examples and are, therefore, exemplary embodiments and not considered to be limiting in scope.

[0011] FIG. 1 illustrates an example environment for a multi-mode rolling-encoded ultrasound device in accordance with one or more implementations.

[0012] FIG. 2 illustrates an example implementation of the ultrasound system from FIG. 1.

[0013] FIG. 3 illustrates a block diagram showing the concept of encoding and decoding signals in a multi-mode rolling-encoded ultrasound.

[0014] FIG. 4 illustrates a block diagram of an example encoding and decoding sequence for two mode types or two variants of the same mode type.

[0015] FIG. 5 illustrates a block diagram of an example encoding and decoding sequence for four mode types or variants of the same mode type.

[0016] FIG. 6 illustrates a block diagram of an example encoding and decoding sequence.

[0017] FIG. 7 illustrates an example of a rolling encoding mechanism in space.

[0018] FIG. 8 illustrates example waveforms for a multi-mode rolling-encoded ultrasound.

[0019] FIG. 9 illustrates additional example waveforms for a multi-mode rolling-encoded ultrasound.

[0020] FIG. 10 outlines a method for a multi-mode rolling-encoded ultrasound.

DETAILED DESCRIPTION

[0021] Conventional ultrasound systems can generate noisy ultrasound images that can result in unideal outcomes and/or failures, such as misclassifications of blood vessels, improper anatomy segmentations, etc. Hence, a patient may not receive the best care possible. Accordingly, systems, devices, and methods are disclosed herein that increase ultrasound resolution without sacrificing frame rate, resulting in a clearer and more accurate ultrasound process compared to conventional ultrasound systems.

[0022] Moreover, an increase in resolution and/or frame rate enables a reduction in undesirable artifacts. Artifacts in ultrasound imagery or processing can present difficulties in identification of anatomy, false positives for present phenomena (e.g., portions of an anatomy, presence of tumors, etc.), or any number of other, unwanted or unideal outcomes from an ultrasound examination. By reducing the number and/or severity of artifacts, a multi-mode rolling-encoded ultrasound enhances the efficacy of an ultrasound examination and related diagnoses.

[0023] In some embodiments, the multi-mode rolling-encoded ultrasound system provides not just ultrasound images but also information related to the examination or examination conditions, such as a speed of sound in the examined medium, a background noise level, a characteristic of the background noise, or other parameters that can be useful in evaluating signals received from an ultrasound examination.

Example System

[0024] FIG. 1 illustrates an example environment 100 for an ultrasound machine 102, in

accordance with one or more implementations. Generally, the ultrasound machine **102** includes various components, some of which include a scanner **104**, one or more processors **106**, a memory **108** storing instructions **110**, an interface module **112**, a communications module **114**, and an output module **116**.

[0025] A user **118** (e.g., nurse, ultrasound technician, clinician, operator, sonographer, etc.) directs the scanner **104** toward a patient **120** to non-invasively scan internal bodily structures (e.g., anatomies, organs, tissues, etc.) of the patient **120** for testing, diagnostic, or therapeutic reasons. In some implementations, the scanner **104** includes an ultrasound transducer array and electronics communicatively coupled to the ultrasound transducer array to transmit ultrasound signals to the patient's anatomy and receive ultrasound signals reflected from the patient's anatomy (e.g., echoes). In some implementations, the scanner **104** is an ultrasound scanner, which can also be referred to as an ultrasound probe or transducer. In aspects, the scanner **104** is configured to produce and receive encoded ultrasound signals, the encoded ultrasound signals comprising two or more ultrasound mode types, two or more variants of a single mode type, or a combination of mode types and mode type variants. For brevity, the present disclosure can refer to all of these as simply “mode types” or “modes.”

[0026] The output module **116** can be, in some examples, a display such as a display **122**. The output module **116** is coupled to the one or more processors **106**, which process the reflected ultrasound signals to generate ultrasound data. The output module **116** is, according to some examples, configured to generate and display an ultrasound image of the anatomy based on the ultrasound data generated by the one or more processors **106** from the reflected ultrasound signals detected by the scanner **104**. In aspects, the ultrasound data includes the ultrasound image or data representing the ultrasound image. In aspects, the output module **116** is configured to generate a non-image output based on the ultrasound data, such as a speed of sound in the medium of the patient **120** anatomy, a background noise, or other non-image information.

[0027] FIG. 2 illustrates an example implementation **200** of the ultrasound machine **102** from FIG. 1. The scanner **104** (e.g., ultrasound scanner) includes an enclosure **202** extending between a distal end portion **204** and a proximal end portion **206**. The enclosure **202** includes a central axis **208** (e.g., longitudinal axis) that intersects the distal end portion **204** and the proximal end portion **206**. The central axis **208** corresponds to an axial direction of the scanner **104**. In an example, the scanner **104** is electrically coupled to an ultrasound imaging system (e.g., the ultrasound machine **102**) via a cable **210** that is attached to the proximal end portion **206** of the scanner **104** by a strain-relief element **212**. In some implementations, the scanner **104** is wirelessly coupled to the ultrasound imaging system and communicates with the ultrasound imaging system via one or more wireless transmitters, receivers, or transceivers over a wireless connection or network (e.g., Bluetooth™, Wi-Fi™, etc.).

[0028] A transducer assembly **214** having one or more transducer elements is electrically coupled to system electronics **216** in the ultrasound machine **102**. In operation, the transducer assembly **214** transmits ultrasound energy from the one or more transducer elements toward a subject and receives ultrasound echoes from the subject. The ultrasound echoes are converted into electrical signals by the transducer element(s) and electrically transmitted to the system electronics **216** in the ultrasound machine **102** for processing and generation of one or more ultrasound images.

[0029] Capturing ultrasound data from a subject using a transducer assembly (e.g., the transducer assembly **214**) generally includes generating ultrasound signals, transmitting ultrasound signals into the subject, and receiving ultrasound signals reflected by the subject. A wide range of frequencies of ultrasound can be used to capture ultrasound data, such as, for example, low-frequency ultrasound (e.g., less than 15 megahertz (MHz)) and/or high-frequency ultrasound (e.g., greater than or equal to 15 MHz). A particular frequency range to use can readily be determined based on various factors (e.g., depth of imaging, desired resolution, etc.).

[0030] In some implementations, the system electronics **216** include one or more processors (e.g.,

the processor(s) **106** from FIG. 1), integrated circuits, application-specific integrated circuits (ASICs), Field Programmable Gate Arrays (FPGAs), Graphics Processing Units (GPUs) and power sources to support functioning of the ultrasound machine **102**. In some implementations, the ultrasound machine **102** also includes an ultrasound control subsystem **218** having one or more processors. At least one processor, FPGA, ASIC, GPU, etc. causes electrical signals to be transmitted to the transducer(s) of the scanner **104** to both emit sound waves and also receive electrical pulses from the scanner **104** that were created from the returning echoes. One or more processors, FPGAs, ASICs, GPUs, etc. process the raw data associated with the received electrical pulses and form an image that is sent to an ultrasound imaging subsystem **220**, which causes the image to be displayed (e.g., via the output module **116**). In aspects, the output module **116** displays ultrasound images from the ultrasound data processed by the processor(s) of the ultrasound control subsystem **218**.

[0031] In some implementations, the ultrasound machine **102** also includes one or more user input devices (e.g., a keyboard, a cursor control device, a microphone, a camera, etc.) that input data and enable taking measurements, such as from a display from output module **116** of the ultrasound machine **102**. The ultrasound machine **102** can also include a disk storage device (e.g., computer-readable storage media such as read-only memory (ROM), a Flash memory, a dynamic random-access memory (DRAM), a NOR memory, a static random-access memory (SRAM), a NAND memory, etc.) for storing the acquired ultrasound data. In aspects, the disk storage device includes the memory **108**, which is local to the ultrasound machine **102**. Alternatively, the memory **108** used for storing the acquisition data can be remote, such as on a remote server communicatively connected to the ultrasound machine **102**. In addition, the ultrasound machine **102** can include a printer that prints the image from the displayed data. To avoid obscuring the techniques described herein, some elements, such user input devices, disk storage device, and printer, are not shown in FIG. 2.

Example Encoding and Decoding

[0032] FIG. 3 illustrates a block diagram **300** showing the concept of encoding and decoding signals in a multi-mode rolling-encoded ultrasound. Diagram **300** illustrates a multi-mode rolling-encoded ultrasound method. There are, in aspects, outgoing modes Mode **1 302**, Mode **2 304**, Mode **3 306**, etc. The Modes **1, 2, 3**, etc. (**302, 304, 306**, etc.) are, according to some examples, transmitted **308** as mixed signals **310**. In aspects, the transmission **308** includes an encoding of the Modes **302, 304**, and **306** into the mixed signal **310** form. In aspects, the mixed signals **310** include waveforms with different polarities (positive or negative) to encode with different patterns.

[0033] In some examples, the encoding of the mixed signals **310** can be performed using an orthogonal operator. For example, each waveform can be first correlated with different, specific mode types (e.g., B-mode, M-mode, A-mode, a Doppler-based mode such as color, PW, Tissue Doppler Imaging (“TDI”), Continuous Wave Doppler (“CW”), etc.) or with the same mode with different delays or apertures. For example, the mixed signals **310** can be encoded with both B-mode and PW mode types. In another example, B-mode imaging with different beam angles or different focal depths or different focal-spot sizes can also be used for encoding the mixed signals **310**. In aspects, the same method can be extended to all the other modes, including M-mode, color, PW, TDI, CW, etc. In some examples, any of the combinations of the above modes and/or different angles, different focal depths, or different focal-spot sizes can be encoded. In general, any combinations in terms of different delays, weights, or waveforms in the transmit events can also, in aspects, be encoded into the mixed signals **310**.

[0034] In aspects, there are several different ways to perform the encoding itself (e.g., Hadamard encoding, Fourier encoding, etc.). In some examples, any orthogonal basis can be used as the encoding mechanism. The following discussion uses Hadamard matrix encoding and decoding as exemplary, but it should be understood that similar formulations with equivalent results can be realized using other encoding and decoding schemes (e.g., Fourier encoding, other orthogonal

matrix formulations, non-orthogonal or non-matrix formulations, etc.). For example, to encode two mode types or two forms of the same mode type, a bipolar Hadamard matrix can be employed of a size two by two:

$$[00001] \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \text{ Matrix1}$$

[0035] Alternatively, a unipolar Hadamard matrix can be employed of a size two by two:

$$[00002] \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \text{ Matrix2}$$

[0036] The negative sign in the bipolar Hadamard matrix of Matrix 1 represents the polarity of the transmit waveform. Note that the negative forms of the Hadamard matrix are mathematically identical in implementation to the positive matrices discussed above. For example, the negative forms of the Hadamard matrix are as follows:

$$[00003] \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} -1 & -1 \\ -1 & 0 \end{bmatrix} \text{ Matrix3\&4}$$

[0037] For simplicity, only the bipolar form of the orthogonal basis for the encode and decode process is described below, but this should not be seen as limiting. Any of the other forms previously disclosed or other orthogonal matrices can be employed to equal effect. For example, the unipolar form applies in a similar manner. It should further be noted that the order of the matrix is not limited to two and can readily be extended to a higher order. In aspects, there are higher order corresponding orthogonal bases and matrices for computation. For example, a bipolar Hadamard matrix of size four by four is:

$$[00004] \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \text{ Matrix5}$$

[0038] Other orthogonal matrices and operators, including non-unitary matrices, matrices with non-zero traces, etc. can be employed to equal effect. The use of a Hadamard matrix is meant to be illustrative and not limiting.

[0039] In some examples, after the mixed signals 310 are transmitted 308, the mixed signals 310 are received 312 as Mode 1 signals 314, Mode 2 signals 316, Mode 3 signals 318, etc. In aspects, because the encoding mechanism is known, the reversed decoding process is used to unmix the mixed signals. In examples using Hadamard encoding, the decoding matrix is the same as the encoding Hadamard matrix.

[0040] FIG. 4 illustrates a block diagram 400 of an example encoding and decoding sequence for two mode types or two variants of the same mode type. The block diagram 400 is an example of bipolar Hadamard encoding and decoding processes for two modes, Mode 1 402A, 402B and Mode 2 404A, 404B. In aspects, the transmission signals 406A and 406B represent Mode 1 based signals (402A and 402B, respectively), and the signals 408A and 408B represent Mode 2 based signals (404A and 404B, respectively). In aspects, transmission signals 406A and 408A are mixed, with both having positive polarity, as mixed signal 410A, and the mixed signal 410A is incident on a target 412A. The target 412A, according to some examples, returns a reception signal 414A, which can be recorded as mixed data 416A. In aspects, the mixed data 416A can be expressed as a combination of received radio frequency (RF) pings RF.sub.406 and RF.sub.408 as:

$$[00005] 416A = RF_{406} + RF_{408} \quad \text{Eq. 1}$$

[0041] In Eq. 1, RF.sub.406 is the RF data corresponding with the Mode 1 402A from the transmission signal 406A and RF.sub.408 is the RF data corresponding with the Mode 2 404A from the transmission signal 408A. Information for both RF.sub.406 and RF.sub.408 are contained in the mixed data 416A.

[0042] In aspects, transmission signals **406B** and **408B** are mixed with positive and negative polarity, respectively, as mixed signal **410B**, and the mixed signal **410B** is incident on a target **412B** (which, in some examples, can be the same target as target **412A**). In aspects, the mixed data **416B** can be expressed as a combination of received RF pings RF.sub.406 and RF.sub.408 as:

$$[00006] \ 416B = RF_{406} - RF_{408} \quad \text{Eq. 2}$$

[0043] In Eq. 2, RF.sub.406 is the RF data corresponding with the Mode **1 402B** from the transmission signal **406B** and RF.sub.408 is the RF data corresponding with the Mode **2 404B** from the transmission signal **408B**. Note that RF ping RF.sub.406 is the same for Mode **1 406A** and Mode **1 406B** and RF ping RF.sub.408 is the same for Mode **2 408A** and Mode **2 408B** as these modes are the same mode type.

[0044] In aspects, the RF data of the Mode **1 (402A and 402B, which is RF.sub.404)**, and the RF data of the Mode **2 (404A and 404B, which is RF.sub.408)** can be easily recovered based on Eqs. 1 and 2 as follows:

$$[00007] \ RF_{406} = (416A + 416B) / 2 \quad \text{Eq. 3} \quad RF_{408} = (416A - 416B) / 2 \quad \text{Eq. 4}$$

[0045] It should be readily apparent that the formulation of Eqs. 3 and 4 is equivalent to the bipolar Hadamard matrix of size 2 by 2, as in Matrix **1**. This follows from the polarities used in mixing the mixed signal **410A** and the mixed signal **410B**.

[0046] In this way, the Hadamard matrix of size 2 by 2, in this example, can be used in both encoding and decoding using either two distinct mode types or two variants of the same mode type. In aspects, the additive operation Eq. 1 can be represented as a data addition **418**, resulting in Mode **1** data **420**, and the negative operation Eq. 2 can be represented as a data subtraction **422**, resulting in Mode **2** data **424**.

[0047] FIG. 5 illustrates a block diagram **500** of an example encoding and decoding sequence for four mode types or variants of the same mode type. Throughout the following discussion, the technique will refer to “four modes,” but it should be understood to include schemes where there is a mix between different mode types and variants within a single mode type, or simply four variants within a mode type in addition to four distinct mode types. To encode and decode four modes, in aspects, a fourth order Hadamard matrix can be used, such as the bipolar Hadamard matrix of Matrix **5**. Mode **1 502A** through Mode **4 508A** can be, in aspects, related to respective transmission signals **510A-516A**. In aspects, the transmission signals **510A-516A** are mixed with positive polarities into a mixed signal **518A** and directed at a target **520A**. The target **520A**, according to some examples, returns a reception signal **522A**, which can be recorded as mixed data **524A**. In aspects, the mixed data **524A** can be expressed as a combination of received RF pings RF.sub.510, RF.sub.512, RF.sub.514, and RF.sub.516 as:

$$[00008] \ 524A = RF_{510} + RF_{512} + RF_{514} + RF_{516} \quad \text{Eq. 5}$$

[0048] In Eq. 5, RF.sub.510 is the RF data corresponding with the Mode **1 502A** from the transmission signal **510A** and RF.sub.512 is the RF data corresponding with the Mode **2 504A** from the transmission signal **512A**. In addition, RF.sub.514 is the RF data corresponding with the Mode **3 506A** from the transmission signal **514A** and RF.sub.516 is the RF data corresponding with the Mode **1 508A** from the transmission signal **516A**.

[0049] Similarly, in aspects, Mode **1 502B** through Mode **4 508B** are related to transmission signals **512B-516B** combined as mixed signals **518B**, which are aimed at a target **520B**, resulting in reception signals **522B** giving mixed data **524B**. According to some examples, the transmission signals **510B-516B** are mixed with polarities corresponding to row two of the bipolar Hadamard matrix of Matrix **5**, allowing the mixed data **524B** to be expressed as a combination of received RF pings RF.sub.510, RF.sub.512, RF.sub.514, and RF.sub.516 as:

$$[00009] \ 524B = RF_{510} - RF_{512} + RF_{514} - RF_{516} \quad \text{Eq. 6}$$

[0050] Similarly, in aspects, Mode **1 502C** through Mode **4 508C** are related to transmission signals **512C-516C** combined as mixed signals **518C**, aimed at a target **520C**, resulting in reception signals

522C giving mixed data **524C**. According to some examples, the transmission signals **510C-516C** are mixed with polarities corresponding to row three of the bipolar Hadamard matrix of Matrix 5, allowing the mixed data **524C** to be expressed as a combination of received RF pings RF.sub.510, RF.sub.512, RF.sub.514, and RF.sub.516 as:

$$[00010] \ 524C = RF_{510} + RF_{512} - RF_{514} - RF_{516} \quad \text{Eq. 7}$$

[0051] Similarly, in aspects, Mode **1502D** through Mode **4 508D** are related to transmission signals **512D-516D** combined as mixed signals **518D**, aimed at a target **520D**, resulting in reception signals **522D** giving mixed data **524D**. According to some examples, the transmission signals **510D-516D** are mixed with polarities corresponding to row four of the bipolar Hadamard matrix of Matrix 5, allowing the mixed data **524D** to be expressed as a combination of received RF pings RF.sub.510, RF.sub.512, RF.sub.514, and RF.sub.516 as:

$$[00011] \ 524D = RF_{510} - RF_{512} - RF_{514} + RF_{516} \quad \text{Eq. 8}$$

[0052] As in the 2 by 2 case shown in FIG. 4, recovery of the RF data for Mode **1 (502A through 502D)** through Mode **4 (508A through 508D)** can be, in aspects, performed using Eqs. 5-8 as follows:

$$[00012] \ RF_{510} = 524A + 524B + 524C + 524D \quad \text{Eq. 9}$$

$$RF_{512} = 524A - 524B + 524C - 524D \quad \text{Eq. 10} \quad RF_{514} = 524A + 524B - 524C - 524D \quad \text{Eq. 11}$$

$$RF_{516} = 524A - 524B - 524C + 524D \quad \text{Eq. 12}$$

[0053] Though, in this example, the magnitude can be increased, this is immaterial to the data recapture as it only represents a scaling factor. It should be readily apparent that this formulation is equivalent to the bipolar Hadamard matrix of size four by four, as in Matrix 5. In this way, the Hadamard matrix of size four by four, in this example, can be used in both encoding and decoding using either distinct mode types, variants of the same Mode type, or a mix of both.

[0054] FIG. 6 illustrates a block diagram **600** of an example encoding and decoding sequence. In aspects, the example of FIG. 4 does not enable an increase over standard ultrasound frame rates as there are still only two data recovered (**420** and **424**) from the initial two transmissions (**410A** and **410B**). To increase frame rate, some examples can include a third transmission signal (or more transmission signals, e.g., four, five, etc.). Considering FIG. 6, in aspects, Modes **1** and **2 (602A through 602C and 604A through 604C, respectively)** are related to transmission signals **606A through 606C and 608A through 608C, respectively**, and combined into mixed signals **610A through 610C, respectively**. The mixed signals **610A through 610C** are aimed at targets **612A through 612C, respectively**, resulting in reception signals **614A through 614C, respectively**, and giving mixed data **616A through 616C, respectively**.

[0055] In aspects, the transmission signals **610A through 610C** can be mixed with different polarities in order to recover received RF pings corresponding to the Modes **1 (602A through 602C)** and **2 (604A through 604C)** via the transmission signals **606A through 606C and 608A through 608C, respectively**, represented as received RF pings RF.sub.606 and RF.sub.608, respectively. In this way, the received mixed data **616A through 616C** can contain information on the RF pings as follows:

$$[00013] \ 616A = RF_{606} + RF_{608} \quad \text{Eq. 13} \quad 616B = RF_{606} - RF_{608} \quad \text{Eq. 14}$$

$$616C = RF_{606} + RF_{608} \quad \text{Eq. 15}$$

[0056] According to some examples, the encoding scheme resulting in Eqs. 13-15 can be used to recover Mode **1** data **620A** and **620B** using addition operators **618A** and **618B** and Mode **2** data **624A** and **624B** using subtraction operators **622A** and **622B** as follows:

$$[00014] \ 620A = 616A + 616B \quad \text{Eq. 17} \quad 620B = 616B + 616C \quad \text{Eq. 18}$$

$$624A = 616A - 616B \quad \text{Eq. 19} \quad 624B = 616B - 616C \quad \text{Eq. 20}$$

[0057] Combining Eqs. 13-15 with Eqs. 16-19 results in the following:

$$[00015] \ 620A = RF_{606} + RF_{608} + RF_{606} - RF_{608} \quad \text{Eq. 20}$$

$$620B = RF_{606} - RF_{608} + RF_{606} + RF_{608} \quad \text{Eq. 21}$$

$$624A = RF_{606} + RF_{608} - RF_{606} + RF_{608} \quad \text{Eq. 22}$$

$$624B = RF_{606} - RF_{608} - RF_{606} - RF_{608} \quad \text{Eq. 23} \quad RF_{606} = \frac{620A}{2} = \frac{620B}{2} \quad \text{Eq. 24}$$

$$RF_{608} = \frac{624A}{2} = -\frac{624B}{2} \quad \text{Eq. 25}$$

[0058] In aspects, the example of FIG. 6 has three outgoing signals in mixed signals **610A**, **610B**, and **610C**, but four received data signals in Mode 1 data **620A** and **620B** and Mode 2 data **624A** and **624B**, resulting in a frame rate increase of 4/3. This gives the following formula for the number of frames M given the outlined rolling encoding/decoding scheme with N input pings:

$$[00016] \quad M = 2N - 2 \quad \text{Eq. 26}$$

[0059] Thus, in aspects, the rolling encoding scheme of FIG. 6 effectively doubles the frame rate, less the first and last pings. In this example, every additional ping over the first ping (and aside from the last ping) generates two additional data. Although two mode types were outlined, this should not be seen as limiting. Similar frame rate gains can be realized using higher order mode type counts, such as three, four, etc.

[0060] In aspects, a potential issue in actual applications is that ping positions cannot be constant, for example in B-mode. In examples of this kind, the correlation among different pings cannot be maintained, causing issues in implementing some or all of the aforementioned encoding/decoding processes. To address this issue, some example embodiments of a multi-mode rolling-encoded ultrasound can include a spatial rolling-encoding and decoding process, as shown in FIG. 7.

[0061] FIG. 7 illustrates an example of a rolling-encoding mechanism **700** in space. In this embodiment, a transducer **702** (e.g., the scanner **104**) can take advantage of a broader transmission and multi-line (ML) reception. For each of these two or more transmissions, in aspects, there can be two or more corresponding receptions and RF data. In some examples, there can be a series of pings sent out with multi-mode encoding. In FIG. 7, dual mode type or mode variant pings are used, though more mode types or variants within a mode type can equivalently be used. For example, a multi-mode ping **704** can be sent with a Mode **704A** and a Mode **704B**. In aspects, a multi-mode ping **706** can be sent with a Mode **706A** and a Mode **706B**, and a multi-mode ping **708** can be sent with a Mode **708A** and a Mode **708B**. Note that Modes **704B** and **706A** overlap in space at line **710** and Modes **706B** and **708A** overlap in space at line **712**.

[0062] Spatial rolling-encoding can be realized in a similar fashion as the multi-mode encoding of previous examples, such as that shown in FIGS. 4, 5, and 6. Again using the non-limiting example of a bipolar Hadamard matrix encoding and decoding, Modes **704A**, **706A**, **708A**, **704B**, **706B**, and **708B** can use multi-mode types. The first mode type can, in some examples, correspond with received RF pings represented by RF.sub.1 and the second mode type can correspond with received RF pings represented by RF.sub.2. The encoding in such an example, when using a bipolar Hadamard matrix encoding, has the following representation:

$$[00017] \quad 704B = RF_1 + RF_2 \quad \text{Eq. 27} \quad 706A = RF_1 - RF_2 \quad \text{Eq. 28}$$

$$706B = RF_1 - RF_2 \quad \text{Eq. 29} \quad 708A = RF_1 + RF_2 \quad \text{Eq. 30}$$

[0063] Reception of these Modes RF.sub.1 and RF.sub.2 can be recovered from signals received as a multi-mode reception signal, such as those originating at the lines **710** and **712**. In aspects, as the line **710** represents a region of space where the Modes **704B** and **706A** are incident and the line **712** represents a region of space where the Modes **706B** and **708A** are incident. For example, taking the polarities of Eqs. 27-30, results in the following equations:

$$[00018] \quad RF_1 = (704B + 706A) / 2 \quad \text{Eq. 31} \quad RF_2 = (704B - 706A) / 2 \quad \text{Eq. 32}$$

$$RF_1 = (706B + 708A) / 2 \quad \text{Eq. 33} \quad RF_2 = (708A - 706B) / 2 \quad \text{Eq. 34}$$

[0064] According to the illustrated example of FIG. 7, Eqs. 31 and 32 are using data from signals originating at the line **710** and Eqs. 33 and 34 are using data from signals originating at the line

712. Thus, in examples and embodiments with spatial rolling encoding and decoding as outlined, the issue of pings originating from variable points in space can be resolved.

[0065] In addition to resolving, in aspects, the spatial encoding problem, the example of spatial rolling encoding and decoding shown by FIG. 7 also can be used to increase the frame rate. If, for example, N is the level of encoding, each ping **704**, **706**, and **708** has two corresponding multi-mode signals with N=2. As detailed by Eqs. 31-34, data recovery here is four frames of data (two of RF.sub.1 and two of RF.sub.2). Each multi-mode signal **706A** and **706B** corresponds with two received data frames, giving the ping **706** four recovered data frames with only two sent signals (the multi-modes **706A** and **706B**). In aspects, the frame rate for such a multi-mode rolling spatial encoding and decoding scheme with multi-modes of order N and outgoing signals of order M gives a frame rate F of:

[00019] $F = M * N$ Eq. 35

[0066] Though the discussion of the multi-mode rolling spatial encoding and decoding scheme of FIG. 7 uses the example of two multi-modes per ping, with each multi-mode consisting of two mode types and using a Hadamard bipolar matrix operation for polarity and recombination, this should not be seen as limiting. Other orders for the pings and mode types can be used, as well as different encoding and decoding operators.

[0067] FIG. 8 illustrates example waveforms for a multi-mode rolling-encoded ultrasound. Waveform **800** is an example of a B-mode waveform. Waveform **802** is an example of a PW waveform. In aspects, the waveforms **802** and **804** represent two different mode types. Other mode types can be used equally, as discussed, but for this example only a B-mode and a PW waveform are used to aid in illustration.

[0068] In a typical ultrasound transducer, the B-mode waveform **800** and the PW waveform **802** can, in aspects, be sent as separate wave pulses or pings. In some examples, a multi-mode rolling-encoded ultrasound can combine the waveforms **800** and **802** into a single ping. Waveform **804** is an example of a B-mode **804-1** and a PW waveform **804-2** in the same ping. Here, the encoding for the waveform **804** uses a positive polarity B-mode **804-1** and a positive polarity PW waveform **804-2**, as in the waveforms **800** and **802**.

[0069] In order to encode with negative signs, as in the examples discussed earlier using bipolar Hadamard matrix encoding and decoding, the polarity of one or more of the mode types encoded into an encoded waveform can be switched. An example of this is waveform **806**. The waveform **806**, in aspects, contains a B-mode **806-1** (as in the waveform **800**) and a PW **806-2** (as in the waveform **802**) waveform, but the PW portion **806-2** has had its polarity flipped. Comparing the PW portion **806-2** of the waveform **806** and the PW portion **804-2** of the waveform **804** illustrates this example polarity reversal. The flipping or reversal of a polarity can, in aspects, correspond with a negative sign in an encoding or decoding operator, such as the negative signs found in a bipolar Hadamard matrix operator.

[0070] FIG. 9 illustrates additional example waveforms for a multi-mode rolling-encoded ultrasound. Instead of using different mode types, waveforms **900-914** represent the same mode type with different variants. In aspects, the waveforms **900-914** can be B-mode waveforms with four distinct angles. The waveforms **900-906** represent single B-mode waveforms, while the waveforms **908-914** represent multi-mode B-mode waveforms. Each of the multi-mode waveforms **908-914**, in aspects, has an encoding for the B-mode varieties included.

[0071] For example, the single-mode waveforms **900-906** can be positive polarity waveforms. Note each of the waveforms **900-906** have the same shape as they are all B-mode waveforms in this example. The multi-mode waveform **908** includes all of the B-mode waveforms **908-1**, **908-2**, **908-3**, and **908-4** from the waveforms **900**, **902**, **904**, and **906** in positive polarity. This is, in aspects, a representation of the first row of a four by four Hadamard matrix, such as the Hadamard matrix of Matrix 5. This can, in aspects, be represented as a vector, (1, 1, 1, 1).

[0072] In another example, the multi-mode waveform **910** includes all of the waveforms **910-1**,

910-2, **910-3**, and **910-4** from the single-mode waveforms **900**, **902**, **904**, and **906**, but the polarities are not all positive. The waveform portions **910-2** and **910-4** both have negative polarities, while the waveform portions **910-1** and **910-3** both have positive polarities. In this example, the waveform **910** follows the second row of the Hadamard matrix of Matrix 5. This can, in aspects, be written as a vector, (1, -1, 1, -1).

[0073] In another example, the multi-mode waveform **912** includes all of the waveforms **912-1**, **912-2**, **912-3**, and **912-4** from the single-mode waveforms **900**, **902**, **904**, and **906**, but the polarities are not all positive. The waveform portions **912-3** and **912-4** both have negative polarities, while the waveform portions **912-1** and **912-2** both have positive polarities. In this example, the waveform **912** follows the third row of the Hadamard matrix of Matrix 5. This can, in aspects, be written as a vector, (1, 1, -1, -1).

[0074] In another example, the multi-mode waveform **914** includes all of the waveforms **914-1**, **914-2**, **914-3**, and **914-4** from the single-mode waveforms **900**, **902**, **904**, and **906**, but the polarities are not all positive. The waveform portions **914-2** and **914-3** both have negative polarities, while the waveform portions **914-1** and **914-4** both have positive polarities. In this example, the waveform **914** follows the fourth row of the Hadamard matrix of Matrix 5. This can, in aspects, be written as a vector, (1, -1, -1, 1).

[0075] In this embodiment, the multi-mode waveforms, via the polarities of their constituent components, represent the Hadamard matrix of Matrix 4, namely:

$$[00020] \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \text{ Matrix6}$$

[0076] In the Hadamard matrix example of Matrix 6, each row **1-4** corresponds to the multi-mode waveforms **908-914**, respectively, and each column **1-4** corresponds to the single-mode waveforms **900-906**, respectively. In this way, encoding and decoding of multi-mode waveforms can be performed as previously outlined. It should be understood that, although this example uses multiple-angle B-mode mode type variants, four by four bipolar Hadamard encoding, and linear encoding (e.g., the multi-mode waveforms **908-914** have one waveform component after another in the encoding and propagation scheme), the techniques disclosed herein are not limited to these examples. Alternate mode type variants can also be used, different encoding and decoding methods (e.g., Fourier matrix, non-orthogonal matrix, etc.) can be implemented, and non-linear waveform combination techniques (e.g., stacked waves, superposition, etc.) can be applied.

Example Method

[0077] FIG. 10 outlines a method **1000** for a multi-mode rolling-encoded ultrasound. The method **1000** can be performed by an ultrasound machine, such as the ultrasound machine **200** described herein. At **1002**, a first ultrasound signal is encoded as a mixed waveform. In aspects, the first ultrasound signal includes a plurality of ultrasound signal modes and is configured for transmission at a subject. According to some examples, the ultrasound signal modes include two or more of an A-mode, a B-mode, an M-mode a Doppler-based mode, a non-image based mode, and/or different variants of these modes. Encoding of the mixed waveform can, in aspects, follow the disclosed methodology used to encode the waveforms **908-914**. The encoding can be accomplished, in some examples, using a bipolar Hadamard matrix, a Fourier matrix, another orthogonal operator, a non-orthogonal encoding operation, etc. In aspects, the mixed waveform can be encoded with linear waveform components, such that each component of the multi-mode signal follows the previous component. Additionally or alternately, the mixed waveform can be encoded in a non-linear fashion, for example using a superposition of constituent waveform components. Other encoding techniques known to a person of ordinary skill in the art can also be employed. In some examples, the encoding of the ultrasound signal includes combining the plurality of ultrasound signal modes

based on at least one of a polarity, a time of transmission, or a location of transmission.

[0078] At **1004**, the first ultrasound signal is generated. The signal generation can be, in some examples, performed with an ultrasound transducer, such as the transducer assembly **214**. In examples where the generation of the first ultrasound signal is intended to be part of an ultrasound examination or scan, the first ultrasound signal can be incident upon an anatomy of a patient or some other structure. In aspects, the first ultrasound signal can propagate in a medium and reflect off the anatomy or the structure being imaged.

[0079] At **1006**, an encoded second ultrasound signal is received. In aspects, the received encoded second ultrasound signal can be based on the first encoded ultrasound signal sent to and reflected from the anatomy or the structure being imaged. Receipt of the encoded second ultrasound signal can be performed by the ultrasound transducer, such as the ultrasound transducer assembly **214**.

[0080] At **1008**, the received encoded second ultrasound signal is decoded into ultrasound data. In examples where the encoding of step **1002** was performed using a bipolar Hadamard matrix, the same bipolar Hadamard matrix can be used for the decoding **1008**. By way of example, the bipolar Hadamard matrix of Matrix 5 can be used. This is meant to be illustrative and not limiting, as there are many other encoding and decoding schemes, which can be employed with similar effect (e.g., unipolar Hadamard matrix operators, Fourier matrix operators, other orthogonal matrix operators, non-orthogonal operators, etc.).

[0081] At **1010**, an output is generated based on the ultrasound data. The output can be an image, such as an image of the anatomy or the structure being imaged. In other examples, the output can be a message to a user, such as a status message, an indication of the presence of a disease or a condition, an indication of a parameter (e.g., a speed of sound in the medium, a background noise level, etc.), or any other message that relays information related to the ultrasound data. In aspects, the mixed waveform can contain single-mode waveforms configured to relay or otherwise generate non-image data related to the anatomy, the structure, the medium, or any other aspect of the ultrasound examination or scan.

[0082] Optionally, at **1012**, a filtered ultrasound data is generated. In aspects, the filtered ultrasound data is generated based on the ultrasound data. At **1014**, an output is generated based on the filtered ultrasound data. In aspects, the output generated at **1014** can be materially similar to the output generated at **1010**, but the output of **1014** is based on the filtered ultrasound signal and not the ultrasound signal.

CONCLUSION

[0083] Embodiments of a multi-mode rolling-encoded ultrasound as described herein are advantageous, as they provide for both an increased resolution and increased frame rate in ultrasound scans, as well as a minimization of artifacts. The techniques of the multi-mode rolling-encoded ultrasound disclosed herein also avoid decorrelation issues often found in spatial ultrasound transducing methods, such as when a transducer generates pings in different areas in space during an ultrasound imaging. Further, the techniques described herein enable non-image outputs to be generated from the multi-mode rolling-encoded ultrasound, such as a determination of a speed of sound in a material, a background noise, or other useful parameters or information, which may not be available in an image output. The multi-mode rolling-encoded ultrasound provide increased scanning efficiency, improved patient experience, higher fidelity scanning outcomes, and similar benefits.

Claims

1. An ultrasound device comprising: an ultrasound scanner configured to generate an ultrasound signal, the ultrasound signal: comprising a plurality of ultrasound signal modes; and configured for transmission by the ultrasound scanner at a subject; one or more processors; and a memory, the memory storing instructions that, when executed by the one or more processors, cause the one or

more processors to encode the ultrasound signal as a mixed waveform based on the plurality of ultrasound signal modes.

2. The ultrasound device of claim 1, wherein the plurality of signal modes include two or more of an amplitude mode (“A-mode”), a brightness mode (“B-mode”), a motion mode (“M-mode”), a Doppler-based mode, and different variants of these modes.
3. The ultrasound device of claim 1, wherein the ultrasound scanner is configured to transmit the encoded ultrasound signal at the subject.
4. The ultrasound device of claim 1, wherein the encoding of the ultrasound signal comprises combining the plurality of ultrasound signal modes based on at least one of a polarity, a time of transmission, or a location of transmission.
5. The ultrasound device of claim 4, wherein the plurality of ultrasound signal modes are combined using an orthogonal matrix operator.
6. The ultrasound device of claim 1, wherein at least one of the plurality of ultrasound signal modes is a non-image-based signal mode.
7. An ultrasound device comprising: an ultrasound scanner, the ultrasound scanner configured to receive an encoded ultrasound signal, the encoded ultrasound signal comprising reflections of a plurality of ultrasound signal modes from a subject; one or more processors; and a memory, the memory storing instructions that, when executed by the one or more processors, cause the one or more processors to decode the received encoded ultrasound signal into ultrasound data comprising the plurality of ultrasound signal modes.
8. The ultrasound device of claim 7, wherein plurality of ultrasound signal modes include two or more of an amplitude mode (“A-mode”), a brightness mode (“B-mode”), a motion mode (“M-mode”), a Doppler-based mode, and different variants of these modes.
9. The ultrasound device of claim 7, wherein the decoding of the received encoded ultrasound signal into the ultrasound data is based on one or more of a polarity, a time of transmission, or a location of transmission for the encoded ultrasound signal.
10. The ultrasound device of claim 7, wherein the instructions further cause the one or more processors to produce an output based on the ultrasound data.
11. The ultrasound device of claim 10, wherein the output is one or more of an image of an anatomy of the subject, a determination of a speed of sound, a determination of a background noise, and a detection of one or more artifacts.
12. The ultrasound device of claim 7, wherein the instructions further cause the one or more processors to generate filtered ultrasound data based on the ultrasound data.
13. The ultrasound device of claim 7, wherein the decoding of the received encoded ultrasound signal into the ultrasound data comprises processing the received encoded ultrasound signal using an orthogonal matrix operator.
14. The ultrasound device of claim 13, wherein the orthogonal matrix operator is based on a Fourier transformation.
15. An ultrasound device comprising: an ultrasound scanner, the ultrasound scanner configured to: generate a first ultrasound signal, the first ultrasound signal: comprising a plurality of ultrasound signal modes; and configured for transmission by the ultrasound scanner at a subject; and receive an encoded ultrasound signal, the encoded ultrasound signal comprising reflections of the plurality of ultrasound signal modes from the subject; one or more processors; and a memory, the memory storing instructions that, when executed by the one or more processors, cause the one or more processors to: encode the first ultrasound signal as a mixed waveform based on the plurality of ultrasound signal modes; and decode the received encoded ultrasound signal into ultrasound data comprising the plurality of ultrasound signal modes.
16. The ultrasound device of claim 15, wherein the plurality of ultrasound signal modes are two or more of an amplitude mode (“A-mode”), a brightness mode (“B-mode”), a motion mode (“M-mode”), a Doppler-based mode, and different variants of these modes.

17. The ultrasound device of claim 14, wherein the encoding of the first ultrasound signal and the decoding of the received encoded ultrasound signal comprise processing the plurality of ultrasound signal modes and the received encoded ultrasound signal, respectively, using an orthogonal matrix operator.

18. The ultrasound device of claim 14, wherein the instructions further cause the one or more processors to: produce, based on the ultrasound signal, filtered ultrasound data; and produce, based on the filtered ultrasound data, an output.

19. The ultrasound device of claim 18, wherein the output is one or more of an image of an anatomy of the subject, a determination of a speed of sound, a determination of a background noise, and a detection of one or more artifacts.

20. The ultrasound device of claim 14, wherein the ultrasound scanner is configured to transmit the encoded ultrasound signal at the subject.
